PUBLIC WHITE PAPER | WP-010

Site Readiness, Installation and Commissioning Framework

A controlled path from qualified customer site to safe, documented and accepted operation

PURPOSE

Define the site information, interfaces, responsibilities, verification and evidence required before a Rapid Organic unit is delivered, energized, commissioned and released for customer operation.

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Executive summary

A successful deployment is an engineered interface between a controlled product and a qualified site. Installation should not begin from an optimistic sales description. It should begin from verified conditions, named responsibilities, approved drawings and a release gate.

Control area	Required position	Why it matters
Site qualification	Actual utilities, space, access, environment and workflow are verified.	Prevents late redesign and unsafe assumptions.
Interface control	Every mechanical, electrical, digital and process boundary has an owner.	Stops gaps between supplier and customer scope.
Readiness gates	Delivery, energization and operation require separate approvals.	Keeps schedule pressure from bypassing controls.
Commissioning evidence	Checks, tests, exceptions and acceptance are recorded.	Creates a defensible operating baseline.
Handover	Training, documents, spares and support routes are complete.	Makes the customer capable without founder dependence.

CORE CONCLUSION

The machine is not ready because it has arrived. It is ready only when the product, site, people and approved operating envelope work together and the evidence is signed.

1. Deployment lifecycle

One accountable deployment plan should link commercial scope to engineering, logistics, construction, commissioning and service. Each gate answers a different question.

Gate	Decision question	Minimum evidence
G0 — opportunity fit	Is the use case within the intended product and feedstock envelope?	Screening record and disqualifiers.
G1 — site qualified	Can the proposed site safely support the unit?	Survey, photos, measurements and utility data.
G2 — design frozen	Are interfaces and responsibilities approved?	Issued drawings, load schedule and scope matrix.
G3 — ship release	Are product, site and receiving logistics ready?	FAT, packing plan and readiness certificate.
G4 — energization	Is installation complete and safe to power?	Inspection, permits and pre-start checklist.
G5 — operation	Has SAT proven required functions at the site?	Signed results and controlled exceptions.
G6 — handover	Can the customer operate and obtain support?	Training, documents, spares and acceptance.

2. Site survey and evidence pack

A survey must describe observed conditions, not assumptions copied from a quotation. Evidence should be dated, traceable and sufficient for remote review.

Survey domain	Capture
Location and access	Address, coordinates, routes, restrictions, delivery window and site contact.
Space and geometry	Measured footprint, clearances, ceiling, doors, turning area and maintenance access.
Structure	Floor construction, level, drainage slope, anchorage conditions and load verification.
Utilities	Electrical service, water, drainage, network, ventilation and backup limitations.
Environment	Indoor/outdoor exposure, temperature, humidity, washdown, dust, salt and pests.
Operations	Feedstock source, storage, movement, staffing, cleaning and output route.
Evidence	Annotated plan, photographs, nameplates, test data, unknowns and responsible verifier.

3. Site-selection disqualifiers

Some conditions should pause or reject deployment until corrected. A marginal site can consume more engineering and service capacity than the sale is worth.

Condition	Required disposition
Uncontrolled feedstock	Define segregation and acceptance process before approval.
No lawful output route	Do not install on the assumption that future product sales will solve disposal.
Insufficient power or protection	Obtain qualified electrical design and confirmed capacity.
Unsafe access or egress	Redesign location, handling route or emergency arrangements.
Unmanaged liquids or drainage	Provide contained, lawful collection and discharge pathway.
Incompatible environment	Use rated enclosure or reject location.
No accountable operator	Name, train and authorize personnel before handover.

COMMERCIAL DISCIPLINE

A site that cannot meet minimum readiness is not a flagship reference site, regardless of customer visibility.

4. Interface responsibility matrix

The contract and installation package should assign every boundary to Rapid Organic, the customer or a qualified third party. “By others” is not a complete responsibility definition.

Interface	Required allocation
Civil / structural	Base, anchorage, penetrations, weather protection and restoration.
Electrical	Supply, disconnect, protection, conductor route, permit and final connection.
Mechanical	Ventilation, ducting, water, drains, piping, guards and insulation.
Material handling	Bins, lifts, loading height, output containers and traffic route.
Digital	Network, accounts, cybersecurity approval, data ownership and remote access.
Compliance	Permits, inspections, environmental route and site-specific procedures.
Acceptance	Witnesses, test material, instruments, criteria and signatories.

5. Layout and maintainability

A layout is acceptable only if the unit can be loaded, cleaned, isolated, serviced and removed without creating new hazards or blocking the customer's operation.

Design check	Acceptance question
Operating clearance	Can routine loading and unloading occur without overreach or conflict?
Service clearance	Can panels, motors, filters and modules be accessed with required tools?
Material flow	Do clean, incoming and processed streams avoid cross-contamination?
People and vehicles	Are pedestrian, forklift and delivery routes separated and visible?
Isolation access	Are disconnects and emergency controls unobstructed and identifiable?
Housekeeping	Can spills, residues and wash water be contained and removed?
Future removal	Can the unit or major modules be replaced without destructive site work?

6. Electrical and control interfaces

Electrical work must be designed and completed by appropriately qualified parties under applicable Canadian and local requirements. Product information must not be mistaken for a site electrical design.

Control	Evidence before energization
Supply characteristics	Voltage, phase, frequency, available capacity and measured condition.
Protection	Disconnect, overcurrent, fault-current rating, grounding/bonding and coordination.
Environment	Approved equipment ratings for moisture, dust, washdown and temperature.
Conductors and route	Approved size, method, support, separation and termination.
Safety circuit	Emergency stop, guards, interlocks and safe-state response verified.
Controls / network	Addressing, access, backups, permissions and offline behavior.
Approval	Applicable permits, inspection and authorized sign-off complete.

BOUNDARY

This framework does not specify wire sizes, protection settings or compliance for a particular installation. Those decisions require site-specific engineering and qualified authority.

7. Ventilation, odour and indoor environment

Odour control cannot be reduced to a marketing promise. Site design should manage normal air, loading events, cleaning and credible abnormal conditions.

Design element	Required definition
Source inventory	Loading point, vessel, output handling, drain, residue and storage.
Air path	Capture point, duct, fan, treatment stage, discharge and make-up air.
Performance basis	Flow, pressure, operating mode, measurement method and acceptance limit.
Discharge location	Separation from doors, intakes, people and sensitive neighbours.
Maintenance	Filter/media access, change interval, waste route and differential indication.
Upset response	Alarm, reduced operation, shutdown, containment and notification.
Verification	Baseline, commissioning observation or measurement, and complaint process.

8. Water, drainage and containment

Every liquid stream should have a defined composition, volume range, containment and authorized destination. A floor drain is not automatically an approved disposal route.

Stream or condition	Control
Process water	Source quality, pressure, backflow protection and metering where relevant.
Condensate / leachate	Expected range, collection capacity, level indication and disposition.
Wash water	Cleaning chemistry, solids capture, volume and approved discharge.
Spill	Bund/curb capacity, compatible materials, kit and reporting route.
Freeze risk	Protection, drainage, heat tracing or seasonal operating restriction.
Sampling	Accessible point, safe method, container and chain of custody.
Records	Volumes, abnormal discharge, collection and service actions.

9. Feedstock receiving and storage

The site must preserve the validated feedstock envelope from receipt to loading. Storage time and contamination can change the material before it reaches the machine.

Control point	Requirement
Source definition	Approved generators, material categories and prohibited inputs.
Inspection	Visual check, contamination response and rejection authority.
Storage	Container, cover, time, temperature, pests, liquid control and cleaning.
Preparation	Sorting, size reduction, mixing and recipe controls where validated.
Traceability	Batch or period identifier, source, mass, time and deviation.
Loading	Safe method, maximum quantity, sequence and operator confirmation.
Exception	Quarantine and disposition for off-specification material.

10. Output handling at the site

Processed material remains controlled until its legal status, quality and destination are established. Installation design must support the approved route without calling the output fertilizer or compost by default.

Area	Required control
Collection	Container type, capacity, change method and prevention of mixing.
Identification	Date/batch, feedstock, status, hold/release and intended route.
Storage	Weather, runoff, dust, pests, temperature and residence time.
Sampling	Representative method, tools, hygiene and laboratory chain of custody.
Release	Named authority and evidence required before use or transfer.
Transport	Packaging, documentation, carrier and receiving acceptance.
Fallback	Lawful disposal route if the material fails requirements.

11. Logistics, delivery and receiving

Transport planning must protect the unit and the site. Delivery failure often begins with unverified weights, dimensions, lifting points or access geometry.

Planning item	Evidence
Shipment data	Packed dimensions, mass, centre of gravity and number of packages.
Route	Bridge/road limits, seasonal constraints, permits and final approach.
Handling	Approved lifting points, equipment capacity, rigging plan and competent crew.
Receiving	Staging area, weather plan, inspection and damage documentation.
Storage before install	Security, moisture, temperature, preservation and battery/consumable care.
Packaging disposition	Reuse, recycling, waste and responsibility.
Escalation	Stop criteria for damage, unsafe handling or site mismatch.

12. Installation quality plan

Installation should use controlled drawings and inspection points. Field adaptation must follow change control so the as-installed baseline remains known.

Inspection point	Required record
Arrival inspection	Identity, condition, completeness and transit damage.
Placement	Location, orientation, level, clearance and anchorage.
Connections	Electrical, mechanical, drainage, ventilation and network completion.
Workmanship	Fasteners, sealing, supports, labels, guards and cleanliness.
Configuration	Serials, software, settings and approved field changes.
Pre-start	Foreign-material check, lubrication, rotation and isolation status.
Punch list	Severity, owner, due date and effect on energization or operation.

13. FAT-to-SAT continuity

The site acceptance test should confirm that the shipped configuration still performs after transport and integration. It should not silently redefine failed factory criteria.

Evidence link	Control
Configuration	Match serial numbers, revision, software and approved deviations to FAT.
Instrument control	Use identified, suitable and calibrated measuring equipment.
Dry checks	I/O, rotation, alarms, interlocks, emergency stop and communication.
Wet / process checks	Use agreed test material and defined operating window.
Performance checks	Measure only claims authorized for site acceptance.
Exceptions	Classify, contain and approve temporary deviation with expiry.
Acceptance	Record raw data, observations, witnesses and final decision.

EVIDENCE RULE

A passed SAT proves the stated checks at that configuration, site and time. It does not establish universal performance across feedstocks or customers.

14. Commissioning sequence and stop rules

Commissioning should progress from safe static checks to limited operation and then to the approved duty. The team must know who can stop work and what evidence permits restart.

Stage	Release condition
Safety briefing	Roles, hazards, emergency actions, boundaries and stop authority understood.
Static verification	Installation complete; permits and critical punch items closed.
Controlled energization	Area controlled; measurements and safe-state behavior confirmed.
Functional test	Sequences, alarms, guards and manual controls verified.
Limited process run	Approved material, reduced exposure and enhanced monitoring.
Duty demonstration	Agreed load and duration with recorded conditions.
Restart after stop	Cause understood, containment complete and authorized release recorded.

15. Training and operational release

Training completion should demonstrate competence for assigned tasks. A signature proving attendance does not authorize troubleshooting beyond the defined role.

Role	Minimum competence evidence
Operator	Pre-use inspection, approved loading, normal control, cleaning and stop response.
Site supervisor	Scheduling, feedstock control, records, deviations and escalation.
Maintenance	Isolation, authorized tasks, instruments, parts and return-to-service.
Environmental contact	Output status, sampling, storage, incidents and approved routes.
IT / security	Accounts, network, access review, backups and incident route.
Management	Commercial boundaries, KPI interpretation and change authorization.
Qualification	Knowledge check plus observed practical task and documented limits.

16. Handover and acceptance package

Handover is complete when the customer has the approved product, site baseline, competence and support pathway—not merely when a commissioning team leaves.

Package component	Minimum content
As-installed file	Drawings, utilities, serials, software, settings and deviations.
Test evidence	FAT, receiving, installation checks, SAT and outstanding limitations.
Operating documents	Manuals, work instructions, cleaning and emergency information.
Maintenance / spares	Schedule, initial kit, storage, reorder and service contacts.
Compliance record	Permits, inspections, approvals and site responsibilities.
Training record	Names, roles, competence, authorization and expiry.
Acceptance certificate	Scope accepted, exclusions, punch items, dates and signatures.

17. First 90 days and replication

The early operating period should verify the handover, close site issues and capture lessons before the same design is repeated elsewhere.

Period	Priority actions
Days 0–7	Enhanced monitoring; confirm operator practice, utilities, alarms and material route.
Days 8–30	Review tickets, housekeeping, consumables, output controls and initial KPI quality.
Days 31–60	Close punch items; verify maintenance and recurring issue containment.
Days 61–90	Customer review; freeze corrected as-installed baseline and lessons learned.
Replication decision	Separate product-standard learning from site-specific variation; update templates under change control.

SCALE RULE

Do not copy an installation detail merely because the first site tolerated it. Replicate only controlled, reviewed and applicable design content.

Conclusion

Rapid Organic deployments should be treated as controlled system integrations. The equipment, utilities, building, feedstock workflow, output route, people and service organization jointly determine whether the site can operate safely and produce credible evidence.

The most useful immediate action is a standard site-readiness pack with objective disqualifiers, a responsibility matrix and separate approvals for shipping, energization, operation and handover. That discipline protects customers while preserving the validity of technical and commercial learning.

DECISION PRINCIPLE

Do not ship into ambiguity. Resolve critical site interfaces and responsibilities before the unit leaves controlled custody.

Scope and limitations

This public framework provides general deployment guidance. It does not replace site-specific engineering, equipment instructions, permits, environmental approvals, occupational health and safety requirements, electrical inspection, fire review, insurance, legal advice or decisions by authorities having jurisdiction.

Selected references

Canadian Centre for Occupational Health and Safety — lockout/tagout

CSA Group — Canadian electrical and machinery standards

ISO 12100 — Safety of machinery risk assessment

ISO 9001 — Quality management systems

Rapid Organic WP-004 — Technical Validation Framework

Rapid Organic WP-006 — Flagship Pilot Design

Rapid Organic WP-007 — Manufacturing Readiness

Rapid Organic WP-009 — Lifecycle Service and Reliability